

Ensemble methods

Tasks (Labs 8-9):

1. Implementation of AdaBoost

- (a) Implement your own version of AdaBoost algorithm using trees as a base classifier – use available implementations of decision trees. Below is a description of AdaBoost algorithm:

Training:

- i. Define initial weights: $w_i = 1/n$, $i = 1, \dots, n$.
- ii. For $k = 1, \dots, B$:
 - A. Build classifier f_k with weights w_i .
 - B. Compute weighted classification error for the k -th classifier:

$$\varepsilon_k = \sum_{i=1}^n I[f_k(x_i) \neq y_i] w_i.$$

- C. If $\varepsilon_k \notin (0, 0.5)$ then handle this case appropriately and state clearly what your implementation should do.
- D. Compute scaling factor:

$$\beta_k = \frac{\varepsilon_k}{1 - \varepsilon_k}.$$

- E. Add a pair (f_k, β_k) to the ensemble.
- F. For $i = 1, \dots, n$:
If $f_k(x_i) = y_i$ (the example is correctly classified), then set $w_i = w_i \beta_k$.
- G. Normalize weights:

$$w_i = \frac{w_i}{\sum_{j=1}^n w_j}.$$

Prediction:

- Prediction for a new observation x :

$$\hat{y}(x) = \arg \max_y \sum_{k=1}^B I[f_k(x) = y] \log \left(\frac{1}{\beta_k} \right).$$

- Weighted voting.
 - Classifiers with smaller errors receive larger weights in the final vote.
- iii. In your implementation in point C of training, define clearly what should happen when $\varepsilon_k = 0$ or when $\varepsilon_k \geq 0.5$. Explain why these cases are problematic for AdaBoost.
 - iv. Compare AdaBoost with base trees of depth 1 and with deeper trees. Comment on how the depth of the base learner influences the results and time of training.
 - v. Theoretical questions:
 - A. Explain why we want a classifier with a smaller ε_k to receive a larger weight in the final prediction.
 - B. Assume that 999 classifiers have error $\varepsilon_k = 0.4$ and one classifier has error $\varepsilon_{1000} = 10^{-8}$. Compute the corresponding voting weights in AdaBoost and comment on whether one almost perfect classifier dominates the final decision.

C. Why are very simple base classifiers, such as decision stumps, often sufficient in AdaBoost?

2. (a) Compare the following ensemble methods:

- Single tree
- Bagging
- Boosting (AdaBoost), your implementation
- Gradient Boosting
- XGBoost
- Random Forest

(b) Consider one real dataset and one artificial dataset. You can choose the real dataset from, e.g., the UCI repository. The real dataset should correspond to a classification problem and should be of moderate size. You may use one of the datasets from the previous labs.

For the artificial dataset:

- Generate $X_1, X_2, \dots, X_{10} \sim N(0, 1)$.
- Denote by $\chi_{10}^2(0.5)$ the median of the chi-squared distribution with 10 degrees of freedom.
- Set $Y = 1$ if

$$\sum_{j=1}^{10} X_j^2 > \chi_{10}^2(0.5)$$

and $Y = -1$ otherwise.

- Generate training data of size 2000 and testing data of size 10000.

- (c) Why do we expect Y to be balanced in this artificial dataset?
- (d) For the artificial dataset, describe the shape of the Bayes decision boundary. Which of the considered methods should be better suited to approximate such a boundary, and why?
- (e) Make plots showing how the **training** and **testing** error change with the number of iterations/trees for boosting, bagging, Random Forest, Gradient Boosting, and XGBoost.
- (f) Compare the methods using the same train/test split for all of them.
- (g) For each method, briefly comment whether the improvement over a single tree seems to come mainly from variance reduction, bias reduction, or both.
- (h) Explain why Random Forest may outperform Bagging, although both methods average many trees.